One flight, one track across a reception gap

Version 4: a rule that keeps one track_id across a long ADS-B gap, what it gains, and why it should not be the default yet

EUROCONTROL Performance Review Unit · 04 November 2026

What we found

Today a reception gap of over 30 minutes ends a track. A flight crossing the Atlantic therefore becomes two tracks and two rows in the flight list, and neither row has both airports. The new rule keeps one `track_id` across such a gap when the evidence at its two edges fits one continuous flight.

- **Flights crossing the European box in one track: 25.3% under the current rule, 68.8% under the new one** (6,450 flights over two days). "Crossing" means exactly one of the two airports lies inside the box.
- All flights in one track: 87.6% to 93.4%.
- **Wrong joins are rare where we can check them:** 2 of 4,670 checkable linked gaps (0.04%), and 0 of 40,975 artificial gaps hiding a real landing. But 74.1% of linked gaps cannot be checked at all, because our truth covers Europe only.
- In a real pipeline run on one airframe in ten, crossing flights with both airports right on one row went from 0 to 80, of 285.

We do not recommend making it the default yet. Two things computed along a track break across a gap, and neither is fixed. A level segment spans a gap on 88.8% of the linked tracks checked (143 of 161). Cumulative distance comes up short across a gap, by a median of 1,482 NM. Switching would also create another silent `track_id` boundary. The rule is in the code, and it is off.

The problem

Take a flight from New York to Frankfurt. Receivers hear it leave New York, lose it over the ocean, and hear it again near Ireland hours later. The current rule sees a gap over 30 minutes and starts a new track. The flight list then says this (an illustration, not a record from the data):

Rule and row	Track	ADEP	ADES
Current rule, first row	first piece	KJFK	none
Current rule, second row	second piece	none	EDDF
New rule, the only row	one track	KJFK	EDDF

Anyone asking how many flights went from New York to Frankfurt counts none under the current rule. Over the two days studied, the 6,450 crossing flights made 12,403 tracks, or 1.92 per flight. In the pipeline sample, 0 of 285 crossing flights had both airports right on one row.

Below, a **hole** is a gap over 30 minutes, and its **edges** are the samples either side of it.

The rule in plain words

A hole stays inside one track only if all of these hold. If a value needed for a check is missing, the check fails and the hole splits, except where the table says otherwise.

Check	Setting	Why
Hole not too long	at most 600 min	Longer holes leave too much room for anything to have happened.
At cruise before the hole	lowest altitude of the last 5 samples, among those within 5 min of the edge, at or above 18,000 ft	An aircraft at altitude is not about to land.
At cruise after the hole	lowest altitude of the first 5 samples, among those within 5 min of the edge, at or above 18,000 ft	The same at the other side.
No descent then climb	not both descending faster than 500 ft/min before and climbing faster than that after; a blank vertical rate does not count against it	That is what a stop in the hole looks like.
Callsign agrees	the last callsign before (at most 60 min old) and the first after (within 10 min) are not both present and different; blank counts as agreeing	A different callsign is a different flight. A blank says nothing.
Implied speed not too slow	at least 400 kt: straight-line distance between the edges over the time between them	A stop uses up time, which lowers this speed.
Implied speed not impossible	at most 750 kt	Faster than this is two aircraft sharing an address, not wind.
Real address	`icao24` not `000000`, `000001` or `ffffff`	These placeholders are shared by many aircraft.

Edge positions are read past blank samples: the last position within 5 min before the hole and the first within 5 min after it, since a position often arrives in a later message than altitude and speed.

Everything else stays as it is. A callsign change and a short gap at low altitude still split. No sample is invented: the hole stays empty and only the identity changes. In the code this is `cruise_linked`, also called A10: the current rule, A9 `debounced`, with this one change. A day's run must read 11 hours back rather than 3, or a track across midnight would publish samples twice, and the code enforces that.

What others do. Nobody publishes a tested method, as far as our literature review, "Link flight pieces only where stops cannot fit", could find. Public practice splits on a time gap of 10 to 30 minutes and mostly stops there. OpenSky's 2021 dataset paper rejoins pieces when a straight extrapolation fits within 0.5 degrees, and GAIA publishes rules for splitting; neither reports how often it links wrongly. We found no study that tests linking against real landings hidden in withheld data, which may be a gap in our search rather than proof that none exists. So this study measures its own error rate.

How we tested it

A **false join** is two different flights of one airframe sharing a track. A landing and a take-off vanish, the flight list pairs an airport from one flight with an airport from another, and nothing in the data shows it. A split, by contrast, is visible and leaves two truthful pieces. So we treat a false join as worse, and unclear evidence means split. That weighting is a project decision, not a result.

The days are 2026-06-14 and 2025-06-14, the only worldwide full days with the tables needed. Truth is Network Manager, the European flight data.

Test	What it does	What should happen	Size
Real holes	Each linked hole against Network Manager, matched on airframe and time and never on callsign, since that would score the callsign check against itself. True link: the hole lies inside one flight. False join: a landing or take-off of that airframe falls inside the hole. Otherwise unknown.	No false joins	18,065 linked, of 228,655 holes

Test	What it does	What should happen	Size
Hidden landing	Two consecutive flights of one airframe. Delete from a cruise point in the first to a cruise point in the second, so a real landing sits inside an artificial hole. The cuts that count sit 30 min or more into cruise, where speed alone cannot give it away.	No joins	40,975
Cut flight	Delete the middle of one continuous flight, for 60 to 360 min.	Rejoined	49,518
Real pipeline day	2026-06-14, one airframe in ten, through the real pipeline under both rules, to the flight list and events.	More flights right, nothing else moved	285 crossing flights

A setting had to meet three limits, pooled over the two days: false joins at most 0.1% of checkable real holes; at most 1% of hidden landings joined, read as a 95% upper bound; and at most 5% where the two flights have the same or a blank callsign, the case the callsign check cannot reject. Among settings within those limits, we took the most crossing flights in one track.

Results

Flights	Number	In one track, current rule	In one track, new rule	Tracks per flight, current	Tracks per flight, new
Crossing the European box	6,450	1,633 (25.3%)	4,439 (68.8%)	1.92	1.31
All others	51,986	49,546 (95.3%)	50,115 (96.4%)	1.03	1.01
All	58,436	51,179 (87.6%)	54,554 (93.4%)	1.12	1.04

The counts include 931 flights with no track at all, which read as not in one track. "In one track" says the flight is not split. It does not say the track holds nothing else: the false-join tests below answer that.

False joins.

Evidence	Result	Reading
Real holes that can be checked	2 false joins of 4,670 (0.04%)	Within the 0.1% limit. 2 on 2025-06-14, 0 on 2026-06-14; 0 had the event within 5 min of an edge, where a timing error could explain it.
Real holes that cannot be checked	13,395 of 18,065 (74.1%)	No evidence either way.
Hidden landing, cuts deep in cruise	0 of 40,975 joined	At most 0.007%, at 95% confidence.
The same, same or blank callsign only	0 of 906 joined	At most 0.33%.
The same, callsign check removed	2 of 40,975 joined	At most 0.015%. So the safety does not rest on the callsign check alone.
Hidden landing, shallow cuts	0 of 38,623 joined	Easy cases, near an airport. Reported, not counted toward the limit.

We also ran 193,469 hidden-landing tests with an edge below the floor and a hole within the longest allowed. With the chosen settings they fail the altitude check by construction, so they say nothing here and are left out above.

How settings trade off. All of these have the speed ratio and time-for-a-stop checks off, which the Choices section covers. The chosen setting is the first row.

Speed floor	Longest hole	Crossing flights in one track	All flights in one track	Real false joins, of checkable	Hidden landings joined	Cut flights not rejoined	Within limits
400 kt	600 min	68.8%	93.4%	2 of 4,670 (0.04%)	0 of 40,975	20.9%	yes

Speed floor	Longest hole	Crossing flights in one track	All flights in one track	Real false joins, of checkable	Hidden landings joined	Cut flights not rejoined	Within limits
400 kt	480 min	68.4%	93.3%	2 of 4,633 (0.04%)	0 of 40,665	20.9%	yes
400 kt	360 min	65.1%	92.9%	1 of 4,382 (0.02%)	0 of 38,379	20.9%	yes
315 kt	600 min	69.2%	93.4%	12 of 4,717 (0.25%)	4 of 40,975	9.2%	no
315 kt	480 min	68.8%	93.4%	4 of 4,672 (0.09%)	2 of 40,665	9.2%	yes
315 kt	360 min	65.5%	93.0%	2 of 4,420 (0.05%)	0 of 38,379	9.2%	yes

Hidden-landing counts differ because only holes the setting allows are counted. The longest hole allowed matters for coverage. The speed floor hardly does on real flights, but it matters for false joins and for cut flights.

What keeps the remaining holes split. These are holes inside the flights scored. On crossing flights, 3,921 are linked and 2,061 stay split. Each is listed under the first check it fails, in the order the code applies them.

First test failed	Crossing flights: holes	of which failed nothing else	Other flights: holes	of which failed nothing else
Below the floor after the hole	1,288	856	544	244
Below the floor before the hole	442	113	357	51
Implied speed too slow, or no position to compute it	315	31	131	7
Hole too long	8	7	5	2
Callsign differs	5	4	3	3
Implied speed impossible	3	3	0	0
All holes left split	**2,061**		**1,040**	

The largest cause is an aircraft that reappears below 18,000 ft: 1,288 holes on crossing flights, 62.5% of those left split, and for 856 of them nothing else fails. Our reading, not measured, is that these flights reappear on approach to their destination. Length is not the problem: 8 holes on crossing flights are too long.

How the method got here

Each row explains a final choice. The comparisons come from earlier runs of the study.

What we found	What it did	What we chose
A position is often blank right at a hole's edge, and the first version split those holes because it could not compute a speed. In the final data, 82,212 of all 228,655 holes (36.0%) use a position read past a blank.	At 25,000 ft, 400 kt and 360 min, crossing flights in one track went from 40.1% to 64.4%, and linked holes from 7,745 to 15,951.	Read positions past blank samples.
All flights moved little (87.6% to 89.6%) because short European flights dominate, while crossing flights went only from 25.3% to 40.1%.	The headline hid the flights this is for.	Choose settings by crossing flights first.

What we found	What it did	What we chose
Lower floors and longer holes let more through.	At 400 kt and 600 min: 67.7% at 25,000 ft, 68.8% at 18,000 ft, 69.9% at 10,000 ft.	18,000 ft and 600 min. See Choices below.
Allowing an edge below the floor when its direction fits one flight (old edge climbing, new edge descending).	Crossing flights 68.8% to 70.5%. But 3 false joins among 251 checkable holes linked this way (1.2%), and overall 5 of 4,921 (0.102%), just over the 0.1% limit. 6 of 193,469 hidden landings joined, 6 of them among the 6,297 with the same or a blank callsign. The 251 is under the 300 the study wanted for a firm figure, so the overall rate is what fails.	Rejected by its own limits. It stays in the code, off.

Two counts say how often a position is read past a blank, and they answer different questions. 82,212 of all 228,655 holes (36.0%) is how common it is among holes, whatever the rule decides, at every length and altitude. 9,232 of 18,065 linked holes (51.1%) is how often a link rests on such a position, among the holes the chosen settings keep. Both count a hole when either edge reads past a blank, and neither counts a hole with no position within 5 min, which cannot link. Of the 2 false joins, 0 used such a position.

Real pipeline day

2026-06-14, one airframe in ten, flights that take off and land on the day.

Flights	Number	Rows per flight, current	Rows per flight, new	On one row, current	On one row, new	Both airports right on one row, current	Both airports right on one row, new
Crossing the European box	285	1.80	1.29	88	204	0	80
of which Europe to outside	156	1.82	1.33	48	107	0	45
of which outside to Europe	129	1.77	1.25	40	97	0	35
Inside Europe	2,672	1.05	1.05	2,533	2,547	1,945	1,952

Crossing flights on one row went from 30.9% to 71.6%. Both airports right on one row went from 0 to 80, which is 28.1% of them. The rule finds no new airports: across every row of the comparison, 0 answers went from none to right. It puts answers on one row that were already on two. The ceiling is detection: 149 of 285 crossing flights (52.3%) have no row naming the airport outside the box, matching flights by callsign. 0 crossing flight gained a row, which we did not look into.

A small loss inside Europe. Of 2,672 flights inside Europe, 15 gained both airports right and 8 lost them. Destination went from right to wrong on 8 and from wrong to right on 8. We do not know why, as these CSVs cannot say. Those 8 flights should be looked at one by one before any promotion.

Nothing else moved. Re-segmenting the sample under both rules, over the whole window a day's run reads, gives 38,383 track boundaries under the current rule and 35,993 under the new one. All 2,390 removed sit at holes the rule links (2,390 there, 0 elsewhere), and 0 were added. The pipeline's 16,481 tracks for the day are exactly those re-segmented tracks. Of 3,493 airport take-offs and landings in APDF, the detector finds 2,913 under the current rule and 2,914 under the new one.

Events that changed, on airframes with a linked track, inside the flights scored:

Event type	Current rule	New rule	Change
first_seen	476	293	-183
last_seen	464	281	-183
top-of-climb	473	290	-183
top-of-descent	473	290	-183
level-end	1,185	1,021	-164
level-start	1,185	1,021	-164
AIBT	117	105	-12
AOBT	122	110	-12
ALDT	172	173	+1
level-off-descent-end	584	585	+1
level-off-descent-start	585	586	+1
xing-fl50	394	395	+1
xing-fl100	396	399	+3
xing-fl245	407	421	+14

Most of this is arithmetic. `first_seen`, `last_seen`, `top-of-climb` and `top-of-descent` each fall by one per joined hole, 183 of them inside the flights scored, out of 980 linked holes in the sample. The level events fall by 164, one per hole inside a level segment, covered in the next section. `xing-fl245` changes by +14. Our reading, not measured, is that a floor below FL245 lets one edge sit below it and the other above, so the hole itself reads as a crossing. The change of +3 at FL100 is not explained that way, as neither edge of a linked hole is that low.

`AOBT` **and** `AIBT` **change by -12 and -12. This is not evidence of glued flights.** The block detector gives a track at most one of each, the first off-block and the last on-block, so two pieces that each had one now have one between them. That is the detector's logic; we did not trace each of these individually. Could an on-block on the older piece and an off-block on the newer one mean the aircraft parked in the hole? To check, all 980 linked holes in the sample were labelled against every Network Manager flight of the airframe, whatever its day. The labels use Network Manager times only. Detected events are counted beside them and never decide a label.

Label	Holes	With a landing or on-block event on the older piece	With a take-off or off-block event on the newer piece	With both
True link	302	62	42	10
False join	0	0	0	0
Unknown	218	23	24	3
Airframe not in Network Manager	460	82	58	18

0 holes are false joins. Such events on both sides of a hole occur on checked true links too (10 of 302), so block events there are not proof of a stop. The event columns are lower bounds: a later piece that starts after midnight is absent because the next day was not built, and an event is counted anywhere on a piece rather than near the hole, so they support nothing beyond that sentence. The 218 unknown and 460 not-in-truth holes stay unchecked, the same limit as in the two-day figures.

What this breaks, and is not fixed

Level segments span holes. On 143 of 161 linked tracks inside the flights scored (88.8%), a level start before a hole pairs with a level end after it: 148 segments containing 164 holes. A user computing level time, or how much of a flight is flown level, gets hours of "level flight" during

which nothing was received. We set the condition before measuring that damage on most linked tracks withdraws the recommendation, and this meets it.

Cumulative distance is short across holes. On 182 of the 182 holes compared, the distance added differs from the straight-line distance between the edges by over 1 NM. The median shortfall is 1,482 NM, and 95% of holes are short by at least 305 NM. All 182 have a blank position on a sample next to the hole, and the distance function reads a blank as zero distance, so the hole adds next to nothing. The stored measurement agrees with this calculation on 161 tracks checked, with 0 off, so this is what would be published. A user reading distance flown from a linked track, or comparing it with a route, sees a long-haul flight far too short, with no warning.

The detectors were not changed in this work, and any fix gets its own `version`.

Limits of the evidence

- **Truth is European, and most linked holes cannot be checked.** 74.1% are unknown. A stop outside Network Manager's area is invisible, so how the rule performs elsewhere is unmeasured.
- **Two days.** 2 false joins, 0 on one day and 2 on the other, are too few to say anything about variation.
- **The two-day measurements re-segment existing tracks, not raw data.** They take `opdi-world/osn_tracks` as given. Only the pipeline day starts from the state vectors.
- **The pipeline day is a 10% sample with few crossing flights,** 285 of them. Its one-row share under the current rule is 30.9%, against 25.3% in the two-day data, which shows how far a sample this size can sit from the larger figure.
- **Flights inside one day only.** Eastbound crossings leaving before midnight UTC drop out, so the sample leans westbound. A flight whose first part belongs to a track of the day before reads as fewer tracks than it has.
- **A flight that never reaches the floor is untested.** It cannot link, and no test covers it.
- **Hard same-callsign cases are few.** There are 906, so the bound is 0.33% overall and much weaker in places: the smallest group (holes of 30-120 min) has 14 tests and a bound of 19%. The 40,975 tests come from at most 38,734 flight pairs, and cuts of one pair are not independent, so the bounds are somewhat too tight.
- **Artificial holes are cleaner than real ones,** and cut flights lean to well-covered flights.
- **Cut flights are often not rejoined: 20.9% at 400 kt, against 9.2% at 315 kt.** Implied speed is the first failed check on 8,155 of the 10,373 misses (78.6%). This barely shows on real flights: going from 315 to 400 kt loses 12.9% of rejoined cut flights but 1.8% of linked real holes. We can only partly explain it. Cut-flight tests are not split by flight type, and 89.0% of flights do not cross the box, so most tests are on shorter flights, where a cut can sit in slower flying or on a less direct path. Tests of 360 min, which only long flights can host, miss 6.7%, which fits. But misses first rise with length, 19.2% at 60 min, 24.3% at 120 and 30.6% at 180, which does not. That explanation is ours and not measured, and the pattern remains unexplained.

Choices made, and what they cost

These are judgement calls, and someone could reasonably go the other way.

Choice	What it costs, and the alternative
Real-hole limit tightened from 1% to 0.1%, after seeing results. The first limit could not separate settings, and we want a wrong join to be rare.	Set after the fact, so it is not an independent test. At 1%, 315 kt and 600 min would pass, with 69.2% of crossing flights and 12 false joins (0.25%).
Floor 18,000 ft, chosen by judgement, not by the selection rule. Applied across floors, the rule would pick 10,000 ft.	At 10,000 ft and 400 kt, staying within the limits also needs a speed ratio check of 0.85: 69.5% of crossing flights, 4 false joins of 4,773 (0.08%). We gave that up for half the false joins and, in our judgement, edges further from busy airspace near airports. Without that check, 10,000 ft gives 69.9% but 6 of 4,817 (0.12%), over the limit.
400 kt and 600 min. The selection rule's pick, and a near tie.	315 kt and 480 min gives 4,436 crossing flights against 4,439, a difference of 3. It has 4 real false joins against 2 and 2 hidden landings joined against 0, but misses 9.2% of cut flights against 20.9%. We preferred fewer false joins.
Time-for-a-stop check off. It asks whether the time left over at a fast ground speed is too short for a stop. A larger allowance is looser.	At 35 min, crossing flights fall to 38.3%, and at 20 min to 31.1%. The safety gain is real but small: false joins 2 to 0, hidden landings unchanged at 0.
Speed ratio check off. It compares implied speed with the aircraft's own speed at the edge.	At 0.85, crossing flights 68.4% and false joins 1, against 68.8% and 2. A small trade either way, and we kept the simpler rule.
Direction-aware edges off.	Gives up 70.5% for 68.8% of crossing flights. See How the method got here.
A blank callsign counted with same callsign in the hard case.	108 of 906 tests (11.9%) are blank. The check cannot reject them, so this is the cautious way, but the group is not purely same-callsign flights.
Any landing or take-off in a hole counts as a false join.	Overstates false joins, if anything.

What is not decided

- **Making it the default.** Not recommended now. The rule ships off and `standard` still points to `debounced`.
- **A new silent `track_id` boundary.** Promotion would add one after those at A8 and A9. A join across it returns nothing rather than an error, and nothing in the published data says which rule produced a row, as `osn_tracks` has no version column. Consumers would have to be told.
- **The two fixes needed first.** Level detection must not pair a start and an end across a hole. Cumulative distance must carry across one. Both are detector changes with their own `version`, and neither is designed yet. After them, the event comparison should be run again.
- **Whether to add an airport test,** asking if any airport the aircraft could have stopped at fits inside the hole. Not built here, because it needs a join outside the rule's current form. Our literature review calls it the most promising and least proven idea. Whether it is needed depends on the unknown holes above.

Reproducing this

Run from the `opdi` repository. `run_track_link_day.py` and the regeneration need the cluster, S3 and Network Manager truth. The pipeline day's tables go under `s3a://eurocontrol/opdi-world/research/a10/`.

```
export PYSPARK_DRIVER_PYTHON=$PWD/.venv310/bin/python
unset PYSPARK_PYTHON
python benchmarks/run_track_link_day.py --day 2026-06-14 --arm debounced --steps 01 02 02a 03 04 04b
python benchmarks/run_track_link_day.py --day 2026-06-14 --arm cruise_linked --steps 02 02a 03 04 04b
python benchmarks/regenerate_track_v4.py --check     # what is stale; no cluster needed
python benchmarks/regenerate_track_v4.py --force     # the two-day runs and the pipeline scoring
python benchmarks/build_track_v4_paper.py            # fills this report; no cluster needed
```

Data. All in `papers/track-construction-v4/data/`. `final_grid/` has the settings search and the headline, `final_direction/` the direction-aware comparison, `stage2/` the pipeline day. `round1`, `round2`, `round3_fl250`, `round3_fl180`, `round3_fl100` and `round4_fl180` are earlier runs, kept to explain the method. They came from earlier code and are not regenerated. The 18,000 ft run among them matches the final one on the chosen settings, which is why floors are compared across them.

Provenance. Each file's script, arguments and commit are in the `_manifest.json` beside it. The final files record commit `1da61a5`. They came from `regenerate_track_v4.py --force` on that commit, and `regenerate_track_v4.py --check` reports all three jobs as current. The pipeline day's temporary tables (state vectors and raw tracks under `research/a10`) were deleted afterwards to return bucket space, so re-scoring the pipeline day needs `run_track_link_day.py` run again first.

Render. `index.qmd` holds this text with literal numbers and the regeneration step, but has not been rendered, because Quarto and R are not available on the machine that built it. `index.html` and `track-construction-v4.pdf` are built from the same text and CSVs by `benchmarks/build_track_v4_paper.py` (the PDF with `--pdf`, using reportlab), not through Quarto.